Prediction

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Compare 16s/ITS versus ME-NMR

```
phyXlong <- phyX %>%
  subset_samples(!is.na(Sample.ID)) %>%
  transform_sample_counts(function(x) x / sum(x)) %>%
  filter_taxa(function(x) sum(x > 0)>10, TRUE) %>%
  psmelt() %>%
  dplyr::select(OTU:TSP, ta1:ta7) %>%
  full_join(MEconc_newlong %>%
              filter(!is.na(Participant)) %>%
              filter(`Blood origen`=='ME') ,
            by = c('Sample.ID' = 'Sample ID', 'Participant', 'Period')) %>%
  filter(!is.na(Abundance)) %>%
  filter(!is.na(logy)) %>%
  filter(!str_detect(met,'EREC')) %>%
  filter(!str_detect(met,'TSP')) %>%
  dplyr::select(OTU:ta7,met,category, type:logy,oldlogy)
```

```
tb_cor <- phyXlong %>%
  group_by(OTU,met,ta1,ta2,ta3,ta4,ta5,ta6,ta7) %>%
  do(cor.test(data = ., ~ logy + Abundance,method = 'spearman') %>% tidy)

tb_lmer <- phyXlong %>%
  # filter(met=='uridine') %>%
  # filter(OTU=='sp1') %>%
  mutate(logabu = log(Abundance+0.0001)) %>%
  group_by(OTU,met,ta1,ta2,ta3,ta4,ta5,ta6,ta7) %>%
  do(lmer(data = ., logy ~  logabu + factor(day) + (1|Participant) + (logabu|Participant)) %>

# tb_lmer %>% filter(str_detect(term,'log')) %>% arrange(p.value)


save(file = 'microbiome_vs_nmr_models.RData', list = c('tb_lmer', 'tb_cor','phyXlong'))
```

All versus all (16s vs NMR)

```
load(file = 'microbiome_vs_nmr_models.RData')

fmrt <- list(

  p.value = colDef(
    cell = function(value) {
      if (is.na(value)) return("")
      formatC(value, format = "e", digits = 2)
    }
  ),

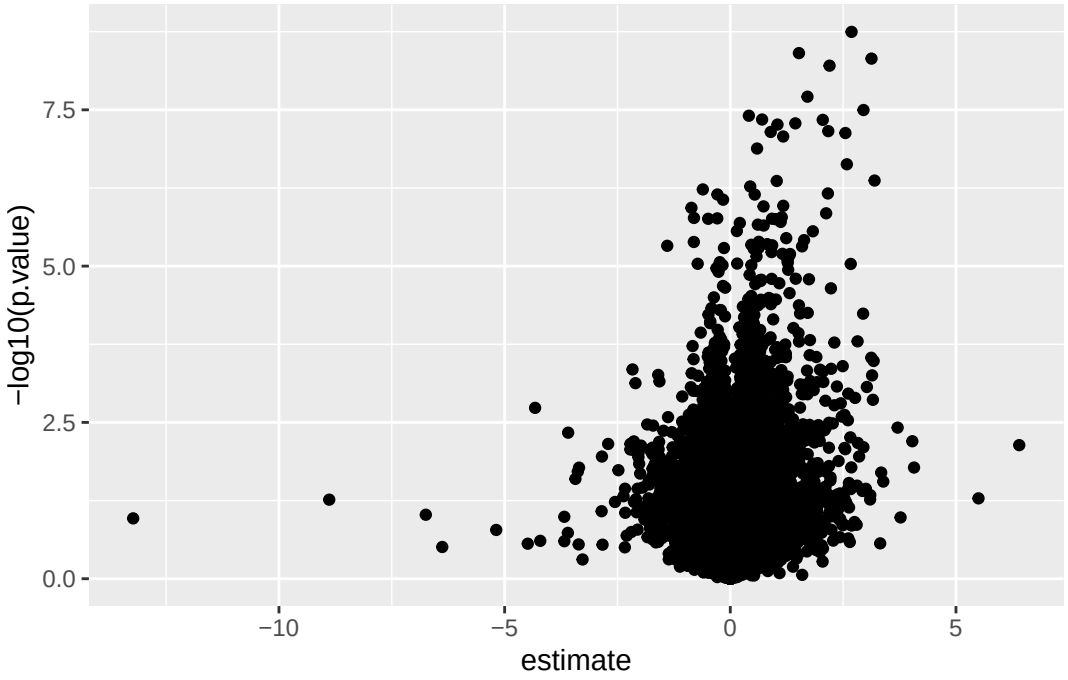
  estimate = colDef(
    cell = function(value) {
      if (is.na(value)) return("")
      formatC(value, format = "e", digits = 2)
    }
  ),

  std.error = colDef(
    cell = function(value) {
      if (is.na(value)) return("")
      formatC(value, format = "e", digits = 2)
```

```
    }
  )
)

tb_lmer2 <- tb_lmer %>%
  filter(term=='logabu') %>%
  ungroup %>%
  mutate(ta5 = ifelse(ta5 %in% c('F__','__'),'Other',ta5),
         ta5lmp = ta5 %>% fct_lump(n = 14),
         qval = p.adjust(p.value,'fdr'),
         pvcut = ifelse(p.value<0.05,'sign','ns'))

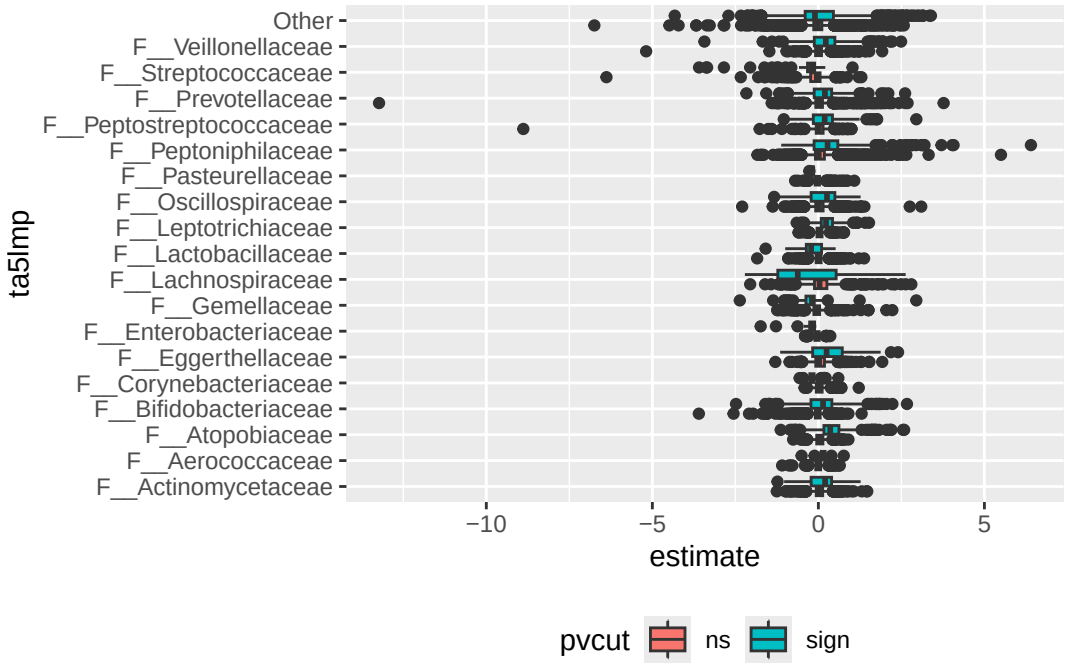
tb_lmer2 %>%
  ggplot(data =.,aes(estimate,-log10(p.value))) +
  geom_point()
```




```
tb_lmer2 %>%
  ungroup %>%
  dplyr::select(met,OTU,ta4,ta5,ta6,ta7,estimate,std.error, p.value) %>%
  reactable(columns = fmrt)
```

met	OTU	ta4	ta5	ta6	ta7	e
2-hydroxybutyric acid	sp1	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	-2
2-hydroxyisovaleric acid	sp1	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	-5
2-oxoisovaleric acid	sp1	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	2
acetate	sp1	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	-2
alanine	sp1	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	3
alpha-glucose	sp1	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	1
aspartate	sp1	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	1
bin-1	sp1	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	-1
bin-1,2-propanediol	sp1	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	7
bin-2	sp1	O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	-6

```
tb_lmer2 %>% filter(str_detect(term,'log')) %>%
  ggplot(data =.,aes(ta5lmp, estimate,fill = pvcut)) +
  geom_boxplot() +
  theme(legend.position = 'bottom') +
  coord_flip()
```

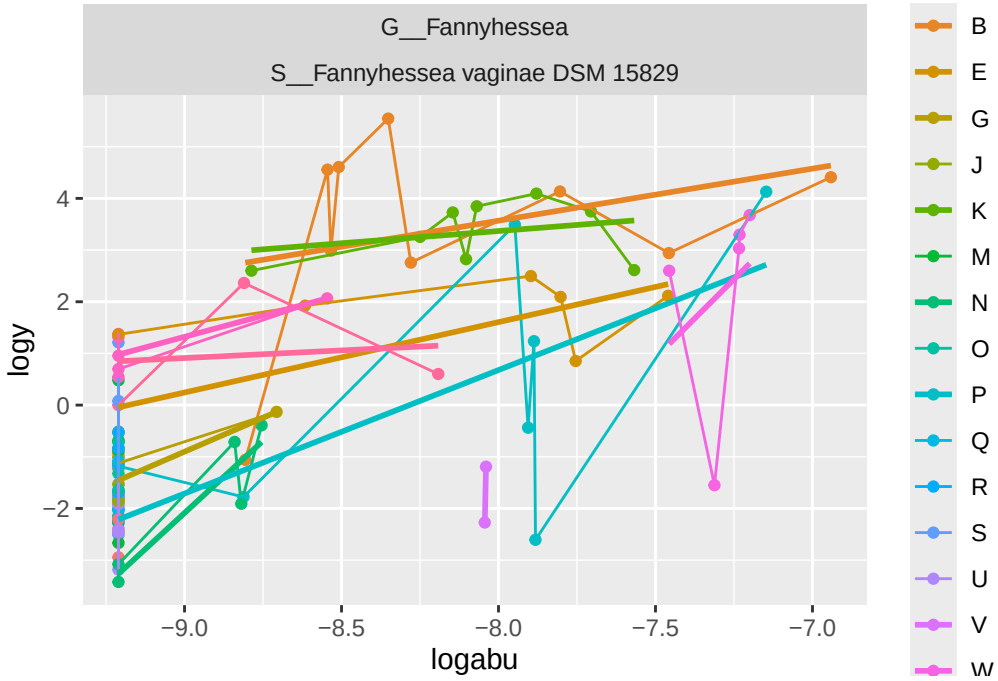



```
# tb_lmer %>% filter(str_detect(term,'log')) %>%
# group_by(ta5lmp) %>%
# dplyr::summarise(nneg = sum(estimate<0), npos = sum(estimate>0),
# pv = 1 - pbinom(max(npos,nneg)-1,nneg+npos,0.5))

gHM <- tb_lmer2 %>%
  group_by(OTU) %>%
  mutate(mq1 = min(qval)) %>%
  ungroup() %>%
  group_by(met) %>%
  mutate(mq2 = min(qval)) %>%
  filter(mq1<0.05 & mq2<0.05) %>%
  mutate(OTU2 = paste(ta4,ta5,ta6,ta7,OTU, sep = '') %>% factor()) %>%
  ggplot(data = ., aes(as.numeric(OTU2),substr(met,1,20),fill = estimate)) +
  geom_tile() +
  scale_fill_gradient2(mid = 'white', midpoint = 0)
```

```
ggsave(filename = 'Met_16s.pdf', plot = gHM, height = 20, width = 20)

phyXlong %>%
  # filter(met=='lactate') %>% filter(OTU=='sp145') %>%
  filter(met=='butyrate') %>% filter(OTU=='sp24') %>%
  mutate(logabu = log(Abundance+0.0001)) %>%
  ggplot(data = ., aes(logabu,logy, color = Participant, group = Participant)) +
  geom_point() + geom_line() + stat_smooth(method = lm, se = F) + facet_wrap(~ta6 + ta7)
```



Multivariate prediciton models using nested-cv

```
library(caret)
library(nestedcv)

make_loio_folds <- function(ids) {
  if (anyNA(ids)) {
    stop("ids must not contain missing values")
  }

  ids <- as.character(ids)
```

```
  n_ids <- length(unique(ids))

  if (n_ids < 3L) {
    stop("At least three unique IDs are required for nested leave-one-ID-out CV")
  }

  # Outer test folds: indices refer to rows in the complete dataset
  outer_folds <- split(
    seq_along(ids),
    factor(ids, levels = unique(ids))
  )

  # Inner test folds: indices refer to rows in each outer-training dataset
  inner_folds <- lapply(outer_folds, function(outer_test) {
    outer_train <- setdiff(seq_along(ids), outer_test)
    train_ids   <- ids[outer_train]

    split(
      seq_along(train_ids),
      factor(train_ids, levels = unique(train_ids))
    )
  })

  list(
    outer_folds = unname(outer_folds),
    inner_folds = lapply(inner_folds, unname)
  )
}
```

From 16s to NMR

```
phyXpp <- phyX %>%
  subset_samples(!is.na(Sample.ID)) %>%
  subset_samples(!is.na(X2.hydroxyisovaleric.acid)) %>%
  transform_sample_counts(function(x) x / sum(x)) %>%
  filter_taxa(function(x) sum(x > 0)>10, TRUE)

X <- phyXpp@otu_table %>% data.frame()
X <- log(X + +0.0001)
Y <- phyXpp@sam_data %>% data.frame() %>% dplyr::select(-TSP,-ERECTIC)
```

```r
folds <- make_loio_folds(Y$Participant)
nmiss <- 1

while (nmiss>0) {
  unmet <- unique(MEconc_newlong$met)
  unmet2 <- unmet <- unmet[!(unmet %in% c("ERECTIC","TSP"))]
  unmet <- unmet %>%
    gsub('-','',x = .) %>%
    gsub(' ','',x = .) %>%
    gsub(',','',x = .)

  fls <- list.files('MLresults16s/') %>% substr(30,100) %>% gsub('_v2.RData','',x = .)
  unmet2 <- unmet2[!(unmet %in% fls)]
  nmiss <- length(unmet2)

  # select a random metabolite from the remaining list
  i <- sample(unmet2)[1]
  print(i)
  flnm <- paste('./MLresults16s/metabolite_16smodels_nestedcv',
                i %>%
                  gsub('-','',x = .) %>%
                  gsub(' ','',x = .) %>%
                  gsub(',','',x = .),
                '_v2.RData', sep = '')
  save(file = flnm, list = c('i'))
  yy <- MEconc_newlong %>% filter(met==i) %>% filter(!is.na(Participant)) %>% filter(`Blood c

  YY <- Y %>%
    left_join(yy, by = c('Participant','Period','Cycle.day' = 'Cycle day'))

  y <- YY$logy
  sdy <- sd(y)

  fit <- nestedcv::nestcv.train(
    x = X,
    y = y,
    method = "spls",                  # any caret model
    preProc = c("center", 'scale'),
    tuneGrid = expand.grid(K = 1:3,# K sets number of components
                           eta = seq(0,1,0.1), # eta sets sparsity
                           kappa = 0.5),
    outer_folds = folds$outer_folds,
```

```
    inner_folds = folds$inner_folds,
    finalCV = FALSE,
    cv.cores = 1
  )

  fit$sdy <- sdy
  fit$metabolite <- i


  save(file = flnm, list = c('fit','i'))
}
```

Permutation null

```
phyXpp <- phyX %>%
  subset_samples(!is.na(Sample.ID)) %>%
  subset_samples(!is.na(X2.hydroxyisovaleric.acid)) %>%
  transform_sample_counts(function(x) x / sum(x)) %>%
  filter_taxa(function(x) sum(x > 0)>10, TRUE)

X <- phyXpp@otu_table %>% data.frame()
X <- log(X + +0.0001)
Y <- phyXpp@sam_data %>% data.frame() %>% dplyr::select(-TSP,-ERECTIC)

c <- 0
CVindex <- list()
for (i in unique(Y$Participant)) {
  c <- c+1
  CVindex[[c]] <- which(Y$Participant!=i)
}


folds <- make_loio_folds(Y$Participant)
nfls <- 1

unmet <- unique(MEconc_newlong$met)
unmet2 <- unmet <- unmet[!(unmet %in% c("ERECTIC","TSP"))]

while (nfls<100) {
  # select a random metabolite from the remaining list
  i <- sample(unmet2)[1]
```

```r
print(i)

yy <- MEconc_newlong %>% filter(met==i) %>% filter(!is.na(Participant)) %>% filter(`Blood c

YY <- Y %>%
  left_join(yy, by = c('Participant','Period','Cycle.day' = 'Cycle day'))

y <- sample(YY$logy) # permute
sdy <- sd(y)

fit <- nestedcv::nestcv.train(
  x = X,
  y = y,
  method = "spls",                    # any caret model
  preProc = c("center", 'scale'),
  tuneGrid = expand.grid(K = 1:3,# K sets number of components
                         eta = seq(0,1,0.1), # eta sets sparsity
                         kappa = 0.5),
  outer_folds = folds$outer_folds,
  inner_folds = folds$inner_folds,
  finalCV = FALSE,
  cv.cores = 1
)

fit$sdy <- sdy
fit$metabolite <- i

yhat <- y
for (ii in 1:18){
  notin <- logical(length(y))
  notin[CVindex[[ii]]] <- TRUE
  # print(mean(y[notin]))
  yhat[!notin] <- mean(y[notin])
}
e <- y - yhat
fit$rmse0 <- caret::RMSE(yhat,y)

summary(fit)
fit$rmse0

nfls <- list.files('MLresults16sperm/') %>% length
flnm <- paste('./MLresults16sperm/metabolite_16smodels_nestedcv_iter',nfls+1,
```

```
                '_v2.RData', sep = '')
  save(file = flnm, list = c('fit'))
}
```

From ITS to NMR

```
phyITSpp <- phyITS %>%
  subset_samples(!is.na(Sample.ID)) %>%
  subset_samples(!is.na(X2.hydroxyisovaleric.acid)) %>%
  transform_sample_counts(function(x) as.numeric(x>0)) %>%
  filter_taxa(function(x) sum(x > 0)>10, TRUE)

X <- phyITSpp@otu_table %>% data.frame()
X <- log(X + +0.0001)
Y <- phyITSpp@sam_data %>% data.frame() %>% dplyr::select(-TSP,-ERECTIC)

folds <- make_loio_folds(Y$Participant)
nmiss <- 1


while (nmiss>0) {
  unmet <- unique(MEconc_newlong$met)
  unmet2 <- unmet <- unmet[!(unmet %in% c("ERECTIC","TSP"))]
  unmet <- unmet %>%
    gsub('-','',x = .) %>%
    gsub(' ','',x = .) %>%
    gsub(',','',x = .)

  fls <- list.files('MLresultsITS/') %>% substr(30,100) %>% gsub('_v2.RData','',x = .)
  unmet2 <- unmet2[!(unmet %in% fls)]
  nmiss <- length(unmet2)

  # select a random metabolite from the remaining list
  i <- sample(unmet2)[1]
  flnm <- paste('./MLresultsITS/metabolite_ITSmodels_nestedcv',
                i %>%
                  gsub('-','',x = .) %>%
                  gsub(' ','',x = .) %>%
                  gsub(',','',x = .),
                '_v2.RData', sep = '')
  save(file = flnm, list = c('i'))
```

```
  yy <- MEconc_newlong %>% filter(met==i) %>% filter(!is.na(Participant)) %>% filter(`Blood c
  YY <- Y %>%
    left_join(yy, by = c('Participant','Period','Cycle.day' = 'Cycle day'))

  y <- YY$logy
  sdy <- sd(y)

  fit <- nestedcv::nestcv.train(
    x = X,
    y = y,
    method = "spls",                  # any caret model
    preProc = c("center", 'scale'),
    tuneGrid = expand.grid(K = 1:2,# K sets number of components
                           eta = seq(0,1,0.1), # eta sets sparsity
                           kappa = 0.5),
    outer_folds = folds$outer_folds,
    inner_folds = folds$inner_folds,
    finalCV = FALSE,
    cv.cores = 1
  )

  fit$sdy <- sdy
  fit$metabolite <- i

  save(file = flnm, list = c('fit','i'))
}
```

Permutation null

```
phyITSpp <- phyITS %>%
  subset_samples(!is.na(Sample.ID)) %>%
  subset_samples(!is.na(X2.hydroxyisovaleric.acid)) %>%
  transform_sample_counts(function(x) as.numeric(x>0)) %>%
  filter_taxa(function(x) sum(x > 0)>10, TRUE)

X <- phyITSpp@otu_table %>% data.frame()
X <- log(X + +0.0001)
Y <- phyITSpp@sam_data %>% data.frame() %>% dplyr::select(-TSP,-ERECTIC)

c <- 0
CVindex <- list()
```

```r
for (i in unique(Y$Participant)) {
  c <- c+1
  CVindex[[c]] <- which(Y$Participant!=i)
}


folds <- make_loio_folds(Y$Participant)
nfls <- 1

unmet <- unique(MEconc_newlong$met)
unmet2 <- unmet <- unmet[!(unmet %in% c("ERECTIC","TSP"))]

while (nfls<100) {
  # select a random metabolite from the remaining list
  i <- sample(unmet2)[12]
  print(i)

  yy <- MEconc_newlong %>% filter(met==i) %>% filter(!is.na(Participant)) %>% filter(`Blood c

  YY <- Y %>%
    left_join(yy, by = c('Participant','Period','Cycle.day' = 'Cycle day'))

  y <- sample(YY$logy) # permute
  sdy <- sd(y)

  fit <- nestedcv::nestcv.train(
    x = X,
    y = y,
    method = "spls",                  # any caret model
    preProc = c("center", 'scale'),
    tuneGrid = expand.grid(K = 1:3,# K sets number of components
                           eta = seq(0,1,0.1), # eta sets sparsity
                           kappa = 0.5),
    outer_folds = folds$outer_folds,
    inner_folds = folds$inner_folds,
    finalCV = FALSE,
    cv.cores = 1
  )

  fit$sdy <- sdy
  fit$metabolite <- i
```

```
  yhat <- y
  for (ii in 1:18){
    notin <- logical(length(y))
    notin[CVindex[[ii]]] <- TRUE
    # print(mean(y[notin]))
    yhat[!notin] <- mean(y[notin])
  }
  e <- y - yhat
  fit$rmse0 <- caret::RMSE(yhat,y)

  nfls <- list.files('MLresultsITSperm/') %>% length
  flnm <- paste('./MLresultsITSperm/metabolite_ITSmodels_nestedcv_iter',nfls+1,
                '_v2.RData', sep = '')
  save(file = flnm, list = c('fit'))
}
```

```
fls <- list.files('MLresults16sperm/')
resperm <- data.frame()
for (fl in fls){
  load(file = paste('./MLresults16sperm/',fl,sep = ''))
  s <- summary(fit)$result %>% as.data.frame() %>%
    rownames_to_column('stat') %>%
    mutate(metabolite = fit$metabolite,method = '16s',sdy = fit$sdy,
           rmse0 = fit$rmse0,
           filename = fl)

  s$x <- as.numeric(s$x)
  resperm <- bind_rows(resperm,s)
}

fls <- list.files('MLresultsITSperm/')
for (fl in fls){
  load(file = paste('./MLresultsITSperm/',fl,sep = ''))
  s <- summary(fit)$result %>% as.data.frame() %>%
    rownames_to_column('stat') %>%
    mutate(metabolite = fit$metabolite,method = 'ITS',sdy = fit$sdy,
           rmse0 = fit$rmse0,
           filename = fl)

  s$x <- as.numeric(s$x)
  resperm <- bind_rows(resperm,s)
}
```

```
save(file = '~/OneDrive - University of Copenhagen/KU_work/menstrual_effluent/multiomics_mens
     list = c('resperm'))
```

```
library(caret)
library(nestedcv)

flnm <- list.files('./MLresults16s/')
beta <- res16s <- data.frame()
for (fl in flnm){
  load(file = paste('./MLresults16s/',fl,sep = ''))
  k <- summary(fit)$final_param$K

  s <- summary(fit)$result %>% as.data.frame() %>%
    rownames_to_column('stat') %>%
    mutate(metabolite = i,method = '16s',sdy = fit$sdy, ncomp = k)
  s$x <- as.numeric(s$x)
  res16s <- bind_rows(res16s,s)


  beta <- beta %>% bind_rows(data.frame(breg = fit$final_fit$finalModel$betahat,
                                        metabolite = i,
                                        method = '16s',
                                        sdy = fit$sdy,
                                        ncomp = k
  ) %>%
    rownames_to_column('OTU'))
}

flnm <- list.files('./MLresultsITS/')
resITS <- data.frame()
for (fl in flnm){
  load(file = paste('./MLresultsITS/',fl,sep = ''))
  k <- summary(fit)$final_param$K
  s <- summary(fit)$result %>% as.data.frame() %>%
    rownames_to_column('stat') %>%
    mutate(metabolite = i,method = 'ITS',sdy = fit$sdy, ncomp = k)
  s$x <- as.numeric(s$x)
  resITS <- bind_rows(resITS,s)

  beta <- beta %>% bind_rows(data.frame(breg = fit$final_fit$finalModel$betahat,
```

```
                                          metabolite = i,
                                          method = 'ITS',
                                          sdy = fit$sdy,
                                          ncomp = k
  ) %>%
    rownames_to_column('OTU'))
}

load(file = '~/OneDrive - University of Copenhagen/KU_work/menstrual_effluent/multiomics_mens

MLstat16s <- RMSE0 %>% left_join(res16s %>% filter(stat=='RMSE') %>% filter(!duplicated(metal
MLstatITS <- RMSE0 %>% left_join(resITS %>% filter(stat=='RMSE') %>% filter(!duplicated(metal

MLstat <- bind_rows(MLstat16s,MLstatITS) %>%
  mutate(stat = 'RMSE',
         ncomp = ifelse(is.na(x),0,ncomp),
         x = ifelse(is.na(x),rmse0,x)) %>%
  left_join(MEclass, by = c('metabolite' = 'Interval_name2')) %>%
  mutate(
    Rsquared = 1 - (x^2 / rmse0^2),
    quality = ifelse(Rsquared < (-0.2),'overfitted','ok'),
    Rsquared = ifelse(Rsquared < (-0.2),-0.2,Rsquared))

beta <- beta %>% left_join(MEclass, by = c('metabolite' = 'Interval_name2'))
save(file = '~/OneDrive - University of Copenhagen/KU_work/menstrual_effluent/multiomics_mens
     list = c('MLstat','res16s','resITS', 'beta'))
```

Estimate FDR based on permutation-null

We have constructed a null distribution using identical ML approach including leave-one-person-out CV on scrambled response variables (metabolite concentration).

We have recorded both the mean-only model (RMSE0) as well as the RMSE from the ML procedure and calculated the observed R2 as listed above.

This R2 null distribution is used to estimate the probability under under no-information of observing a value of R2* or larger. Due to limited observations in the tails under permutation, this part of the distribution is estimated using a generalized pareto distribution with a tail from the 0.7 fractile and up.

This smooth functional form of the distribution is used to calculate the mass under null and contrasting this with the actual fraction of discoveries as a empirical false discovery rate.

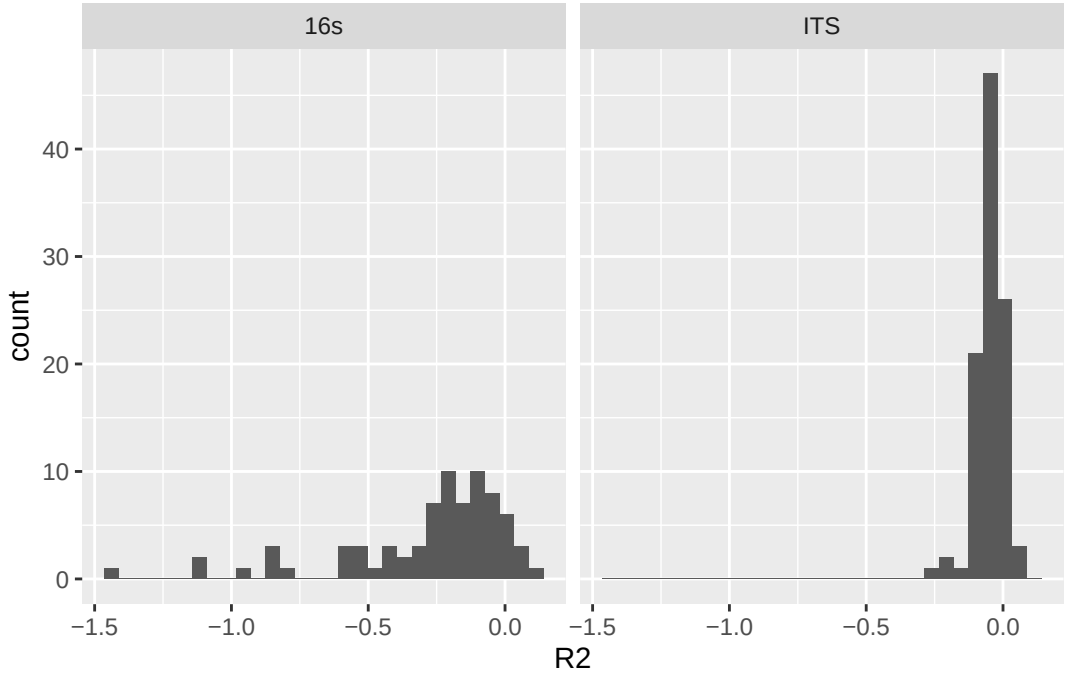
```r
load(file = '~/OneDrive - University of Copenhagen/KU_work/menstrual_effluent/multiomics_mens

load(file = '~/OneDrive - University of Copenhagen/KU_work/menstrual_effluent/multiomics_mens

library(mev)

res <- resperm %>%
  filter(stat=='RMSE') %>%
  mutate(R2 = 1 - x^2 / rmse0^2) %>%
  filter(!duplicated(paste(x,metabolite))) %>%
  group_by(method) %>%
  arrange(R2) %>%
  mutate(nn = 1,
         cdf = cumsum(nn) / sum(nn))

res %>% ggplot(data = ., aes(R2)) + geom_histogram() + facet_wrap(~method)
```




```r
DFprob <- data.frame()
for (mth in unique(res$method)){
  MLstatsel <- MLstat %>% filter(method==mth)
  ressel <- res %>% filter(method==mth)
  data_points <- ressel$R2
```

```
  threshold_val <- quantile(data_points, 0.70)

  gpd_fit <- fit.gpd(xdat = data_points,
                     threshold = threshold_val,
                     method = "Grimshaw")

  # --- 2. Extract the Parameter Estimates ---
  sigma <- gpd_fit$estimate[1] %>% as.numeric() # Scale parameter
  xi    <- gpd_fit$estimate[2] %>% as.numeric()  # Shape parameter
  u     <- gpd_fit$threshold %>% as.numeric()     # Threshold value

  # Calculate empirical probability of exceeding the threshold (zeta)
  n_total     <- length(data_points)
  n_exceed    <- sum(data_points > u)
  zeta_u      <- n_exceed / n_total

  # Using mev's built-in GPD distribution function (pgp)
  dfprob <- data.frame(x = seq(u,0.5,0.001))
  dfprob2 <- data.frame(x = seq(min(data_points),u,0.001))
  dfprob$condp <- pgp(dfprob$x, loc = u, scale = sigma, shape = xi)
  dfprob$uncondp <- (1 - zeta_u) + zeta_u * dfprob$condp
  dfprob <- bind_rows(dfprob,dfprob2) %>% mutate(method = mth)

  dfprob$cdf <- NA

  nmet <- dim(MLstatsel)[1]
  for (i in 1:dim(dfprob)[1]){
    dfprob$cdf[i] <- ressel$cdf[max(which(ressel$R2<dfprob$x[i]))]
    dfprob$frac_disc[i] <- sum(MLstatsel$Rsquared>dfprob$x[i]) / nmet
  }

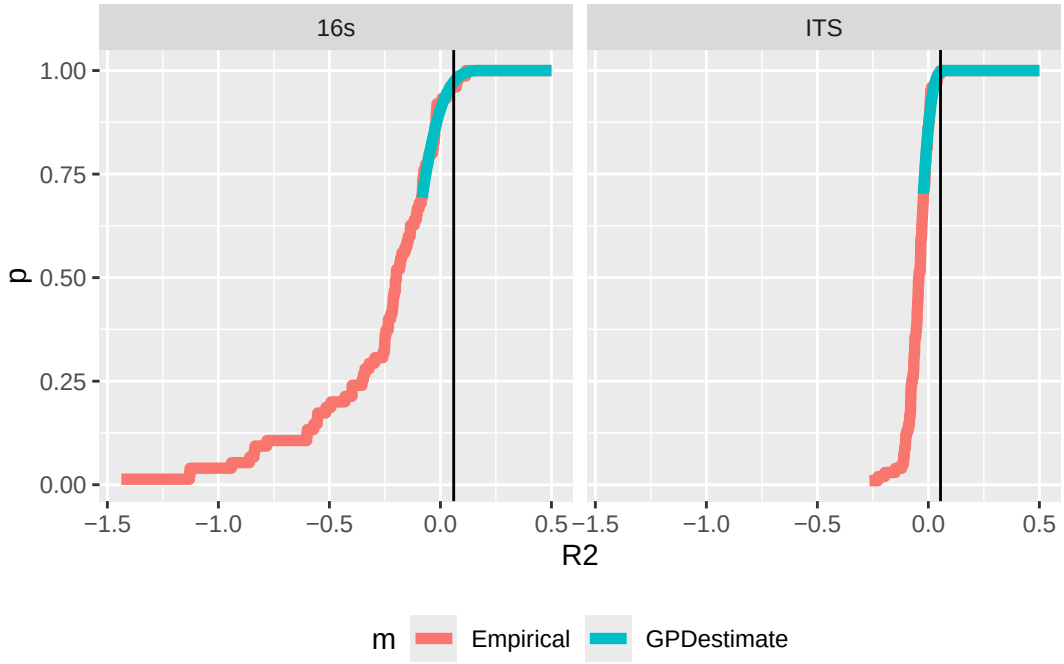
  DFprob <- bind_rows(DFprob,dfprob)
}

DFprob <- DFprob %>%
  # for each value of R2 (in increasing order calculate the FDR value as the propotion of dis
  mutate(fdr = (1-uncondp) / frac_disc)

dffdr <- DFprob %>%
  group_by(method) %>%
  filter(!is.na(fdr)) %>%
  filter(fdr<0.05) %>%
```

```
  arrange(x) %>%
  slice(1)

DFprob %>%
  group_by(method) %>%
  gather(m,p,cdf,uncondp) %>%
  mutate(m = m %>% factor(labels = c('Empirical','GPDestimate'))) %>%
  ggplot(data = ., aes(x,p,color = m)) +
  geom_line(size = 2) +
  facet_wrap(~method) +
  geom_vline(data = dffdr, aes(xintercept = x)) +
  xlab('R2') + theme(legend.position = 'bottom')
```



Per metabolite - CV predictive perormance.

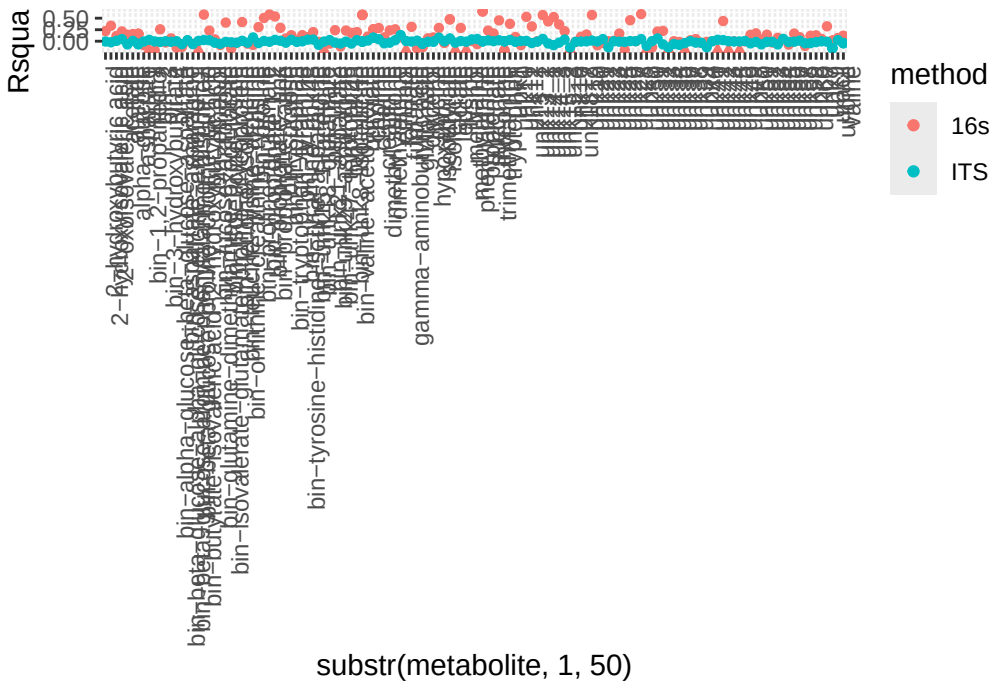
Number of components selected

```
load(file = '~/OneDrive - University of Copenhagen/KU_work/menstrual_effluent/multiomics_mens

table(MLstat$method,MLstat$ncomp)
```

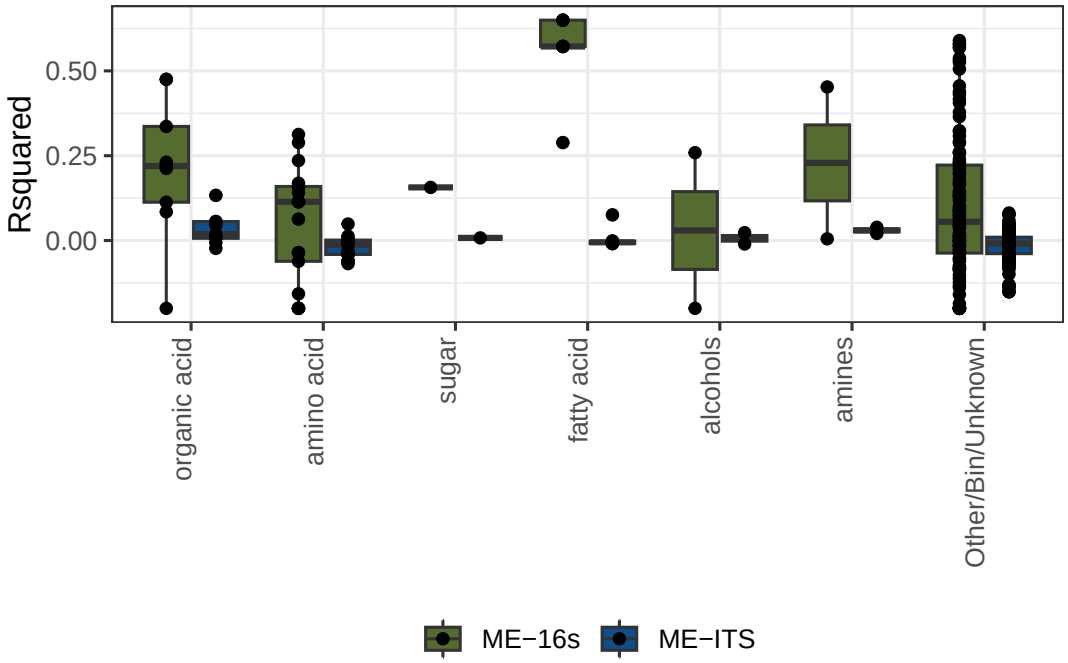
```
       0   1   2   3
16s    6 107  32  11
ITS    3 137  16   0
```

```
MLstat %>%
  ggplot(data = ., aes(substr(metabolite,1,50),Rsquared, color = method, shape = quality)) +
  geom_point() + theme(axis.text.x = element_text(angle = 90, hjust = 1))
```




```
g_r2 <- MLstat %>%
  left_join(dflb, by = 'category') %>%
  filter(!is.na(lb)) %>%
  # group_by(lb) %>%
  # mutate(category2 = paste(lb, ' (#', n() / 2,')', sep = ''))  %>%
  # ungroup %>%
  mutate(lb = lb %>% factor(levels = dflb$lb[1:9])) %>%
  ggplot(data= ., aes(lb,Rsquared, fill = method)) +
  geom_boxplot() +
  # scale_y_reverse() +
  geom_point(position = position_dodge(width = 0.75)) +
  # geom_hline(data = data.frame(R2cut = c(R2sign_16s,R2sign_ITS), method = c('16s','ITS')),
  # aes(yintercept = R2cut, color = method)) +
```

```
  # geom_point(position = position_dodge(width = 0.8), aes(group = method), size = 1) +
  scale_fill_manual(values = colorcodes[1:2], labels = colorcodesnames[1:2]) +
  scale_color_manual(values = colorcodes[1:2], labels = colorcodesnames[1:2], guide = 'none') +
  # scale_x_discrete(breaks = dflb$lb[1:9], values = dflb$lb[1:9]) +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1, vjust =  0),
        legend.position = 'bottom') + labs(fill = NULL) +
  xlab(NULL)
g_r2
```




```
MLstat %>%
  filter(!is.na(Rsquared)) %>%
  lm(rank(Rsquared) ~ method  + category ,data = .) %>% anova %>% tidy
```

```
# A tibble: 3 x 6
  term         df    sumsq  meansq statistic  p.value
  <chr>     <int>    <dbl>   <dbl>     <dbl>    <dbl>
1 method        1  245168. 245168.      37.0   3.69e-9
2 category     14  322781.  23056.       3.48  2.84e-5
3 Residuals   296 1962685.   6631.      NA    NA
```

```
MLstat %>%
  filter(!is.na(Rsquared)) %>%
  group_by(method) %>%
  summarise(R2med = median(Rsquared),
            q1 = quantile(Rsquared,0.25),
            q3 = quantile(Rsquared,0.75))
```

```
# A tibble: 2 x 4
  method    R2med      q1      q3
  <chr>     <dbl>   <dbl>   <dbl>
1 16s     0.0858  -0.0371 0.244
2 ITS    -0.00770 -0.0359 0.00967
```

Figure 5A in publication

```
library(gplots)

ggbeta <- beta %>%
  left_join(MLstat %>% dplyr::select(rmse0,x,Rsquared, metabolite,method,ncomp), by = c('meta
  left_join(dflb, by = 'category') %>%
  mutate(breg = breg/sdy) %>%
  filter(method == '16s') %>%
  filter(Rsquared>0.45) %>%
  left_join(tb_lmer2 %>% dplyr::select(OTU,met,estimate:pvcut), by = c('OTU','metabolite' =
  group_by(metabolite) %>%
  mutate(nn = sum(abs(breg)>0)) %>%
  filter(nn>0) %>%
  group_by(OTU) %>%
  mutate(nn2 = sum(abs(breg)>0)) %>%
  filter(nn2>0) %>%
  left_join(phyX@tax_table %>% data.frame() %>% rownames_to_column('OTU') %>%
              left_join(bsum16s, by = c('ta2','ta3','ta4','ta5')), by = 'OTU') %>%
  ungroup %>%
  mutate(taxall = paste(ta2,ta3,ta4,ta5, sep = '') %>% factor() %>% as.numeric(),
         OTU = OTU %>% fct_reorder(taxall)) %>%
  mutate(stars = gtools::stars.pval(qval))

lbs <- ggbeta %>%
  group_by(OTU) %>%
  dplyr::summarise(lb = ta5[1],
                   lb4 = ta4[1],
```

```
                    lb3 = ta3[1],
                    lb2 = ta2[1],
                    n()) %>%
  arrange(lb)

# 1) find lowest level of id, 2) remove F__
for (i in which(nchar(lbs$lb)<4)){
  id <- which(nchar(lbs[i,])>4)[1]
  if (is.na(id)) {
    lbs$lb[i] <- 'unassigned'
  }
  else {
    # print(as.character(lbs[i,id]))
    lbs$lb[i] <- as.character(lbs[i,id])
  }
}

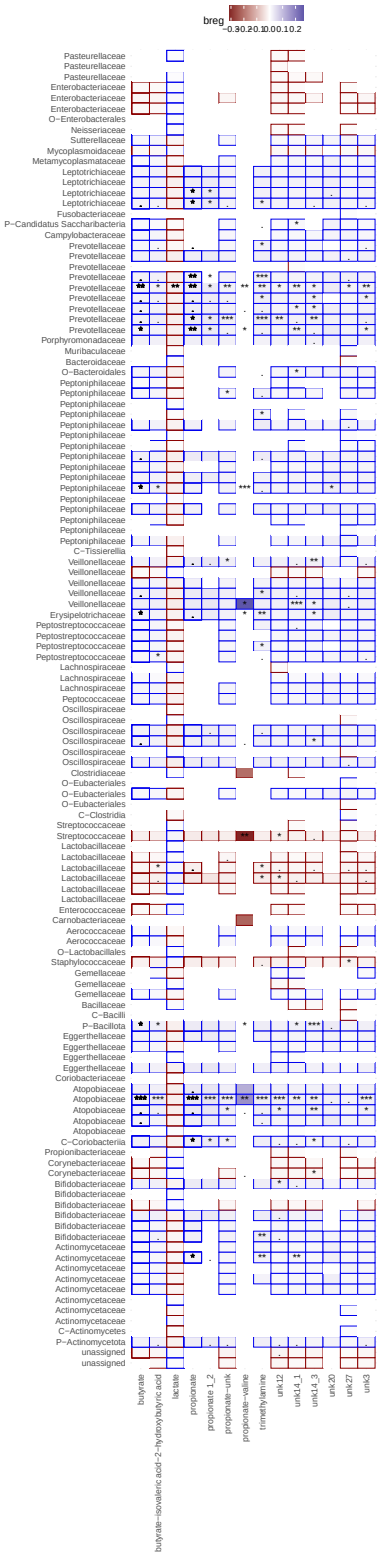
lbs <- lbs %>%
  mutate(lb = lb %>%
           gsub('F__','',x = .) %>%
           gsub('__','-',x = .))


g5_all <- ggbeta %>%
  # remove bin from mtb label
  mutate(metabolite = gsub('bin.','',x = metabolite)) %>%
  # filter of only bins / unk
  # filter(nchar(metabolite)>6) %>%
  mutate(metabolite = gsub('X2.','2-',x = metabolite)) %>%
  # filter(!str_detect(metabolite,'unk')) %>%
  # filter(!str_detect(metabolite,'1_2')) %>%
  ggplot(data = ., aes(metabolite,OTU,fill = breg,color = factor(sign(breg)), label = stars))
  geom_tile() +
  geom_text(color = 'black') +
  scale_y_discrete(labels = lbs$lb,breaks =lbs$OTU ) +
  scale_fill_gradient2(mid = 'white',midpoint = 0) +
  scale_color_manual(values = c('red4',"#FFFFFF00",'blue2'), guide = "none") +
  theme_minimal() +
  theme(legend.position = 'top', axis.text.x = element_text(angle = 90, hjust = 1, vjust = 0
  xlab(NULL) + ylab(NULL)
```

```
g5_sel <- ggbeta %>%
  # remove bin from mtb label
  mutate(metabolite = gsub('bin-','',x = metabolite)) %>%
  mutate(metabolite = gsub('1_2','',x = metabolite) %>% trimws()) %>%
  # filter of only bins / unk
  filter(nchar(metabolite)>6) %>%
  filter(!str_detect(metabolite,'unk')) %>%
  group_by(OTU) %>%
  mutate(nactive = sum(abs(breg))) %>%
  ungroup %>%
  filter(nactive>0.0001) %>%
  # group_by(metabolite) %>%
  # mutate(breg = breg / max(abs(breg))) %>%
  # filter(!str_detect(metabolite,'1_2')) %>%
  ggplot(data = ., aes(metabolite,OTU,fill = breg,color = factor(sign(breg)), label = stars))
  geom_tile() +
  scale_y_discrete(labels = lbs$lb,breaks =lbs$OTU ) +
  scale_color_manual(values = c('red4',"#FFFFFF00",'blue2'), guide = "none") +
  scale_fill_gradient2(mid = 'white',midpoint = 0) +
  geom_text(color = 'black') +
  # scale_color_manual(values = alpha(c('white','black'))) + #,c(0,1))) +
  theme_minimal() +
  theme(legend.position = 'top', axis.text.x = element_text(angle = 90, hjust = 1, vjust = 0
  # guides(color = guide_legend(NULL))+
  xlab(NULL) + ylab(NULL)
```

Coefficients - all

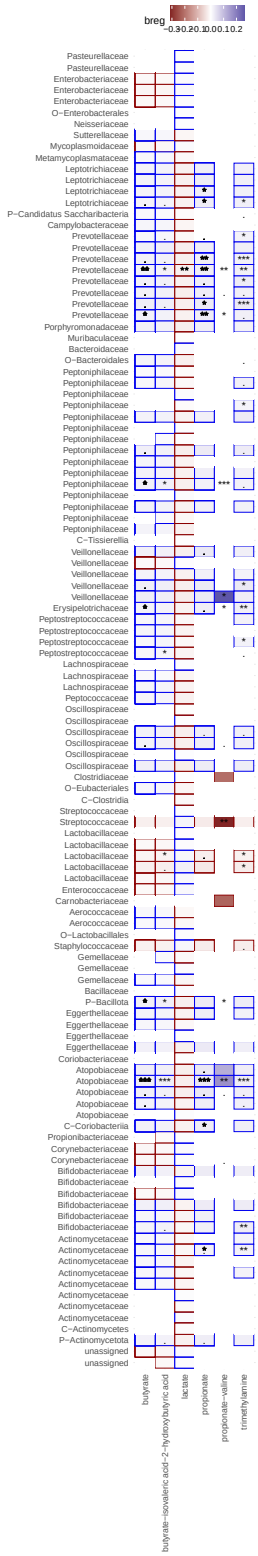
```
g5_all
```


breg
−0.3 −0.2 −0.1 0.0 0.1 0.2
Pasteurellaceae
Pasteurellaceae
Pasteurellaceae
Enterobacteriaceae
Enterobacteriaceae
Enterobacteriaceae
O−Enterobacterales
Neisseriaceae
Sutterellaceae
Mycoplasmoidaceae
Metamycoplasmataceae
Leptotrichiaceae
Leptotrichiaceae
Leptotrichiaceae
Leptotrichiaceae
Fusobacteriaceae
P−Candidatus Saccharibacteria
Campylobacteraceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Porphyromonadaceae
Muribaculaceae
Bacteroidaceae
O−Bacteroidales
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
C−Tissierella
Veillonellaceae
Veillonellaceae
Veillonellaceae
Veillonellaceae
Veillonellaceae
Erysipelotrichaceae
Peptostreptococcaceae
Peptostreptococcaceae
Peptostreptococcaceae
Peptostreptococcaceae
Lachnospiraceae
Lachnospiraceae
Lachnospiraceae
Peptococcaceae
Oscillospiraceae
Oscillospiraceae
Oscillospiraceae
Oscillospiraceae
Oscillospiraceae
Oscillospiraceae
Clostridiaceae
O−Eubacteriales
O−Eubacteriales
O−Eubacteriales
C−Clostridia
Streptococcaceae
Streptococcaceae
Lactobacillaceae
Lactobacillaceae
Lactobacillaceae
Lactobacillaceae
Lactobacillaceae
Lactobacillaceae
Enterococcaceae
Carnobacteriaceae
Aerococcaceae
Aerococcaceae
O−Lactobacillales
Staphylococcaceae
Gemellaceae
Gemellaceae
Gemellaceae
Bacillaceae
C−Bacilli
P−Bacillota
Eggerthellaceae
Eggerthellaceae
Eggerthellaceae
Eggerthellaceae
Coriobacteriaceae
Atopobiaceae
Atopobiaceae
Atopobiaceae
Atopobiaceae
Atopobiaceae
C−Coriobacteriia
Propionibacteriaceae
Corynebacteriaceae
Corynebacteriaceae
Bifidobacteriaceae
Bifidobacteriaceae
Bifidobacteriaceae
Bifidobacteriaceae
Bifidobacteriaceae
Bifidobacteriaceae
Actinomycetaceae
Actinomycetaceae
Actinomycetaceae
Actinomycetaceae
Actinomycetaceae
Actinomycetaceae
Actinomycetaceae
Actinomycetaceae
Actinomycetaceae
C−Actinomycetes
P−Actinomycetota
unassigned
unassigned
butyrate
butyrate−isovaleric acid−2−hydroxybutyric acid
lactate
propionate
propionate_1_2
propionate−unk
propionate−valine
trimethylamine
unk12
unk14_1
unk14_3
unk20
unk27
unk3

Coefficients - selected

Figure 5B in publication

```
g5_sel
```


breg
−0.3 −0.2 −0.1 0.0 0.1 0.2
Pasteurellaceae
Pasteurellaceae
Enterobacteriaceae
Enterobacteriaceae
Enterobacteriaceae
O−Enterobacterales
Neisseriaceae
Sutterellaceae
Mycoplasmoidaceae
Metamycoplasmataceae
Leptotrichiaceae
Leptotrichiaceae
Leptotrichiaceae
Leptotrichiaceae
P−Candidatus Saccharibacteria
Campylobacteraceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Prevotellaceae
Porphyromonadaceae
Muribaculaceae
Bacteroidaceae
O−Bacteroidales
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
Peptoniphilaceae
C−Tissierellia
Veillonellaceae
Veillonellaceae
Veillonellaceae
Veillonellaceae
Veillonellaceae
Erysipelotrichaceae
Peptostreptococcaceae
Peptostreptococcaceae
Peptostreptococcaceae
Peptostreptococcaceae
Lachnospiraceae
Lachnospiraceae
Peptococcaceae
Oscillospiraceae
Oscillospiraceae
Oscillospiraceae
Oscillospiraceae
Oscillospiraceae
Oscillospiraceae
Clostridiaceae
O−Eubacteriales
C−Clostridia
Streptococcaceae
Streptococcaceae
Lactobacillaceae
Lactobacillaceae
Lactobacillaceae
Lactobacillaceae
Lactobacillaceae
Enterococcaceae
Carnobacteriaceae
Aerococcaceae
Aerococcaceae
O−Lactobacillales
Staphylococcaceae
Gemellaceae
Gemellaceae
Gemellaceae
Bacillaceae
P−Bacillota
Eggerthellaceae
Eggerthellaceae
Eggerthellaceae
Eggerthellaceae
Coriobacteriaceae
Atopobiaceae
Atopobiaceae
Atopobiaceae
Atopobiaceae
Atopobiaceae
C−Coriobacteriia
Propionibacteriaceae
Corynebacteriaceae
Corynebacteriaceae
Bifidobacteriaceae
Bifidobacteriaceae
Bifidobacteriaceae
Bifidobacteriaceae
Bifidobacteriaceae
Bifidobacteriaceae
Actinomycetaceae
Actinomycetaceae
Actinomycetaceae
Actinomycetaceae
Actinomycetaceae
Actinomycetaceae
Actinomycetaceae
C−Actinomycetes
P−Actinomycetota
unassigned
unassigned
butyrate
butyrate−isovaleric acid−2−hydroxybutyric acid
lactate
propionate
propionate−valine
trimethylamine